Measurement, Observables, and the Quantum Toolbox

EVA: Quantum Machine Learning – Lecture 3

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Roadmap

1. Projective measurement and shot statistics
2. Observables and expectation values
3. Qiskit primitives: sampler vs estimator
4. Complex numbers in quantum and ML
5. Real-data mini experiment (Iris, two classes)

Measurement and shots

Projective measurement in the computational basis

For a single qubit

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle\,, \qquad |\alpha|^2 + |\beta|^2 = 1,$$

measurement in the Z basis returns:

$$P(0) = |\alpha|^2, \qquad P(1) = |\beta|^2.$$

After measurement, the state collapses to the observed basis state. A single shot gives one random bit; many shots estimate probabilities.

Why we need multiple shots

Let $p = P(1)$ and $\hat{p} = k/N$ from N shots.

$$\mathbb{E}[\hat{p}] = p, \qquad \mathrm{Var}(\hat{p}) = \frac{p(1-p)}{N}.$$

Typical standard error:

$$\sigma_{\hat{p}} \approx \sqrt{\frac{p(1-p)}{N}}$$

scales as $1/\sqrt{N}$.

`lecture_03_demo.ipynb`, Demo 1: shot convergence.

Hoeffding bound for shot planning

For bounded outcomes in $[0, 1]$:

$$\Pr\left(|\hat{p} - p| \geq \epsilon\right) \leq 2e^{-2N\epsilon^2}.$$

To guarantee error ϵ with confidence $1 - \delta$:

$$N \geq \frac{\ln(2/\delta)}{2\epsilon^2}.$$

Example: $\epsilon = 0.01$, $\delta = 0.05$

$$N \gtrsim 18{,}445.$$

This is a worst-case bound; practical circuits may need fewer shots.

Observables and expectation values

What is an observable?

An **observable** is a Hermitian operator O:

$$O = O^\dagger.$$

Hermitian $\Rightarrow$ eigenvalues are real.

In the projective measurement model:

- the measurement outcome is an eigenvalue λ_i
- the post-measurement state is the corresponding eigenvector

Single-qubit observables: matrices I, X, Y, Z

The standard one-qubit Pauli observables (and identity) are

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

For X, Y, Z, the only possible outcomes are ± 1.

What does it mean to "measure I"?

I has only one eigenvalue:

$$I\,|\psi\rangle = |\psi\rangle \quad \Rightarrow \quad \text{outcome is always } +1.$$

Therefore,

$$\langle I\rangle = \langle\psi|\, I\, |\psi\rangle = 1$$

for any normalized state.

Why it still appears: many Hamiltonians and cost functions include constant terms (offsets) written as a multiple of I.

Spectral decomposition (why eigenvectors appear)

Any Hermitian observable O has an orthonormal eigenbasis $\{(\lambda_i, |\phi_i\rangle)\}$ and can be written as

$$O = \sum_i \lambda_i \, |\phi_i\rangle \, \langle\phi_i| \, .$$

Reminder: $\langle\psi| = (|\psi\rangle)^\dagger$ (conjugate transpose).

Interpreting a measurement of O:

- the reported outcome is one of the eigenvalues λ_i
- the post-measurement state is the corresponding eigenvector $|\phi_i\rangle$

Born rule as an overlap (projection)

If the system is in state $|\psi\rangle$, then

$$P(\lambda_i) = |\langle\phi_i|\psi|\phi_i|\psi\rangle|^2 = \langle\psi|\underbrace{|\phi_i\rangle\langle\phi_i|}_{P_i}|\psi\rangle\,.$$

For real unit vectors, $|\langle\phi_i|\psi|\phi_i|\psi\rangle| = \cos\theta$, so $P(\lambda_i) = \cos^2\theta$.

`lecture_03_demo.ipynb`, Demo 1b: overlaps, probabilities, basis change.

Expectation values (definition)

For state $|\psi\rangle$ and observable O:

$$\langle O \rangle = \langle \psi | \, O \, | \psi \rangle \, .$$

If O has eigenpairs $(\lambda_i, |\phi_i\rangle)$, then

$$\langle O \rangle = \sum_i \lambda_i \, P(\lambda_i), \qquad P(\lambda_i) = |\langle \phi_i | \psi | \phi_i | \psi \rangle|^2.$$

Expectation values (statistical meaning)

One shot gives one random eigenvalue. Repeating many shots estimates the mean $\langle O \rangle$.

Example (measuring Z): outcomes are ± 1, hence

$$\langle Z \rangle = (+1)P(0) + (-1)P(1) = P(0) - P(1).$$

In variational QML, expectation values are typical loss-function inputs.

Worked example: $\langle Z \otimes I \rangle$ on a Bell state

Let $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$.

Apply Z to the *first* qubit:

$$(Z \otimes I)\,|00\rangle = |00\rangle\,, \qquad (Z \otimes I)\,|11\rangle = -\,|11\rangle\,.$$

Therefore,

$$(Z \otimes I)\,\left|\Phi^+\right\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}} = \left|\Phi^-\right\rangle.$$

Since $\langle \Phi^+ | \Phi^- \rangle = 0$,

$$\langle Z \otimes I \rangle = 0.$$

In vector/matrix form (basis $|00\rangle\,, |01\rangle\,, |10\rangle\,, |11\rangle$):

$$\left|\Phi^+\right\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \quad Z \otimes I = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

Then

$$(Z \otimes I)\,\left|\Phi^+\right\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix},$$

and the inner product with $|\Phi^+\rangle$ is zero.

What does $\langle Z \otimes I \rangle = 0$ mean?

$\langle Z \otimes I \rangle$ is the expectation of measuring Z on qubit 0 *ignoring* qubit 1.

For $|\Phi^+\rangle$, measuring the first qubit in the Z basis gives

$$P(0) = \frac{1}{2}, \qquad P(1) = \frac{1}{2},$$

hence

$$\langle Z \otimes I \rangle = P(0) - P(1) = 0.$$

Interpretation: each *local* qubit looks unbiased, while the pair can still have strong *correlations* (e.g., $\langle Z \otimes Z \rangle = 1$).

Why we measure local terms like $Z \otimes I$:

- local marginals: behavior of one subsystem only
- Hamiltonians often include one-body terms (Z_i) and two-body terms
- diagnostics: compare local signal ($Z \otimes I$) vs correlation ($Z \otimes Z$)

Bell-state correlations

For

$$\left|\Phi^+\right\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle),$$

key expectation values:

$$\langle Z \otimes Z\rangle = +1, \qquad \langle X \otimes X\rangle = +1, \qquad \langle Z \otimes I\rangle = 0.$$

Reading:

- perfect joint correlation in Z and X
- each local marginal is unbiased

`lecture_03_demo.ipynb`, Demo 2: Bell observables.

Sampler vs estimator

Sampler, Estimator, and Aer: side-by-side

	StatevectorSampler	StatevectorEstimator	AerSimulator / Aer wrappers
Main question	bitstring outcomes?	mean value of O?	same questions, more realism/performance
Circuit has measurements?	yes	no	depends on task
Needs observable?	no	yes	for estimator-style tasks, yes
Typical output	counts / probabilities	$\langle O \rangle$	counts or $\langle O \rangle$
Typical use	classical readout results	VQE/VQC objectives	larger/noisy simulation studies

See also: `lectures/lecture_03/qiskit_primitives_guide.md`.

Which one should I use?

- Use **StatevectorSampler** when your circuit ends with measurements and you care about bitstring statistics.
- Use **StatevectorEstimator** when your target is an expectation value $\langle O \rangle$ for one or more observables.
- Use **AerSimulator** (or Aer-based primitives) when you need:
  - higher-performance simulation for bigger circuits, or
  - realistic noise modeling (for example density-matrix methods).

Same state, two different questions

```
# Sampler question: what outcomes occur?
qc = QuantumCircuit(1)
qc.ry(theta, 0)
qc.measure_all()

# Estimator question: what is <Z>?
qc_no_meas = QuantumCircuit(1)
qc_no_meas.ry(theta, 0)
```

`lecture_03_demo.ipynb`, Demo 3: sampler/estimator comparison.

Complex numbers in quantum and ML

Why complex numbers are operational, not decorative

Complex amplitudes carry phase information that controls interference.

Two states can share the same Z-basis probabilities but differ in phase, yielding different values of $\langle X \rangle$ or $\langle Y \rangle$.

This is why phase-sensitive observables matter in quantum modeling.

Connection to machine learning

Classical models usually use real-valued features and weights.

Quantum states are naturally complex-valued feature vectors in Hilbert space.

Practical bridge used in this course:

- map real data into quantum states
- extract expectation-value features (e.g., $\langle X \rangle, \langle Y \rangle, \langle Z \rangle$)
- train a classical model on these features

Real-data mini experiment (Iris)

In this lecture, we use a two-class Iris subset and compare:

1. baseline logistic regression on raw standardized features
2. quantum-feature baseline from angle encoding with observables $\langle X \rangle, \langle Y \rangle, \langle Z \rangle$

Goal: not advantage claims, but pipeline literacy from real data to quantum-style feature extraction.

`lecture_03_demo.ipynb`, Demo 4: Iris complex/quantum features.

Summary

- measurement is probabilistic, expectation values are statistical means
- shot count controls estimation precision
- observables define what quantity is being measured
- sampler and estimator answer different questions
- real datasets can already be integrated at the measurement stage

Next class: classical ML refresher and kernel methods, then quantum feature maps.

References i

1. Nielsen and Chuang, *Quantum Computation and Quantum Information*.
2. Schuld and Petruccione, *Supervised Learning with Quantum Computers*.
3. Qiskit documentation: `https://docs.quantum.ibm.com/`.
4. PennyLane documentation: `https://docs.pennylane.ai/`.